

Jyandeep Baishya

6002590335 | craftzyan@gmail.com | LinkedIn | GitHub | LeetCode

Experience

Spiral Kite Labs — Software Engineer

Oct 2025 – Present

LangGraph, Next.js, TypeScript, FastAPI, S3, ChromaDB

Lexo

- Unified **30** workbook actions spanning grids, formulas, charts and mindmaps into a shared mutation layer consumed by **Redux**, **backend services** and **LangGraph agents**, eliminating duplicated update paths and **preserving deterministic state transitions** across every execution environment.
- Re-architected workbook synchronization** from snapshot persistence to event-style mutation propagation, reducing average save payloads from **25KB to 211 bytes (121× reduction)** by replaying immutable actions against a versioned in-memory state manager that batches **persistence every 5 seconds**, supports **autosave** semantics, and retains up to 50 active workbooks through **LRU eviction**.
- Established a **schema-driven charting subsystem** covering **12 visualization types** whose configurations persist as versioned workbook state, paired with an **AI-assisted recommendation** step that profiles tabular context and proposes chart mappings through **Pydantic**-validated schemas.
- Prototyped a **NotebookLM-inspired knowledge exploration MindMap** built on an **O(N) layout engine**, zoomable canvas, and **LangGraph multi-agent** topic generation with **tool calling**, letting users navigate workbook content as interactive mindmaps rather than static spreadsheets.
- Exposed workbook capabilities through an **authenticated SSE-enabled CLI** that executes **agent tool calls** against the same workbook engine used by the browser and backend, providing a reproducible environment for iterating on **LangGraph** workflows outside the UI.

ImaginePDF

- Established **pdfTreejs as the canonical document contract** across a normalized tree of **10 node types**, stable identities, and variable overlays, so the editor, API automation layer, and Python renderer could evolve without schema drift.
- Introduced **Canva-style editing** over the document tree — **layers**, **multi-selection**, **atomic undo/redo snapshots**, **rich text**, **tables**, **QR/barcodes**, and group transforms — while preserving WYSIWYG parity through a points-only document model.
- Operationalized a **rendering pipeline** that compiles document trees to **HTML/CSS Paged Media**, generates PDFs via **WeasyPrint**, rasterizes previews via **PyMuPDF**, and returns artifacts through **API-signed URLs**.
- Enabled **document automation** through variable-bound templates, **CSV datasets**, and plugin actions, letting users batch-generate personalized invoices, certificates, and reports while preserving design-defined layouts.
- Extended batch generation through **asynchronous operator workflows** that render one PDF per dataset row, allowing users to export **100s** of personalized documents as a single downloadable artifact, and expose authenticated download surfaces for long-running export jobs.

Twvios Technology — Founding Engineer

Mar 2025 – Oct 2025

React, Next.js, Redux Toolkit, TailwindCSS

- Co-owned the architecture of lastminutesdeal.com, a multi-domain marketplace spanning **customer**, **seller** and **admin** experiences — designing **catalog/inventory** models, **search & discovery**, **cart**, **checkout**, **payments**, **auth**, **couponing**, **order management** and **booking workflows** with role-based dashboards; drove **150+** responsive **Next.js** interfaces alongside a 5+ intern team while standardizing reusable components and frontend delivery workflows.

Technical Skills

AI Systems: Python, LangGraph, LangChain, RAG, ReAct, Multi-Agent Systems, MCP, Tool Calling, Evals, LLMOPS

Development: TypeScript, Node.js, Express.js, FastAPI, REST APIs, React, Next.js, Redux Toolkit, TailwindCSS, shadcn/ui

Education

National Institute of Technology Silchar

2020 – 2024

B.Tech, Electrical Engineering

CGPA: 7.1

Projects

Agentic RAG Assistant | LangGraph, FastAPI, Elasticsearch, Redis, SSE

- Orchestrated a **multi-agent chat system** on a **LangGraph StateGraph** — a **supervisor** routing each turn across **3 specialized agents** (search, summary, parallel) via reducer-based shared state — replacing brittle if/else flow with a **checkpointed, replayable** graph persisted per thread in **Redis**, enabling durable memory that survives restarts plus **cross-session recall**.
- Engineered **hybrid Retrieval-Augmented Generation** in **Elasticsearch** fusing **BM25** and **dense-vector kNN** retrieval to lift answer relevance **~25%** over keyword-only search, and streamed tokens over **SSE** to cut **time-to-first-token ~8× (2.5s → <300ms)**; **capability-based model routing** right-sized each call to reduce **LLM cost ~30%**.
- Hardened a **FastAPI** backend with **JWT** auth, **60 req/min** rate limiting, and a **Redis** response cache cutting repeat-query latency **~50%**; added **human-in-the-loop** approval via **LangGraph interrupt/resume**, **tool-calling**, and a **50+ case** eval suite running in CI.